

Problem Statement and Proposed Solution

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1. The problem

Citizens of Madhya Pradesh already have ways to complain: the CM Helpline, a toll-free number, email, WhatsApp and the Bhopal Municipal Corporation (BMC) portal. What they lack is a way to **describe a problem in their own words and have it reach the right office without knowing how the government is organised**.

Evidence from Madhya Pradesh (Bhopal is our first example; the same pattern applies in every district)

Finding	Source	Confidence
BMC is the civic body for Bhopal; head office at Harshwardhan Complex, Mata Mandir, Bhopal 462001	bhopal.nic.in	High, official
Civic complaints (water, road, sewerage, streetlight) can be lodged through the CM Helpline, a toll-free number, email, WhatsApp or the BMC portal, and a complaint needs a zone or ward number	complainthub.org	Medium, unofficial guide
The structure is department → body → zone → ward. Ward count is quoted as 85 or 86 and zone count as 19 or 21 by different unofficial sources	<code>docs/research/T05-research.md</code>	Unverified, conflicting
Who owns a "no water" complaint is ambiguous: BMC distribution versus PHED for bulk supply	<code>docs/ARCHITECTURE_DEPARTMENTS.md</code>	Ambiguous
MPOne's own Customer Solutions Hub is a voice/call-centre-first platform with an executive dashboard; multilingual support is listed only as a future item	eservicesnest.com (competitor research, <code>docs/COMPETITOR_RESEARCH_ESERVICES_NEST.md</code>)	Their own marketing claims, not independently verified

What this means for the citizen

1. **Knowledge barrier.** The citizen must know the department and the ward or zone before filing.
2. **Language barrier.** Forms and menus assume standard written language; many people speak Hindi, Hinglish or a dialect.
3. **Misrouting.** Ownership is genuinely ambiguous in places, so complaints land in the wrong queue.
4. **No feedback.** Without a complaint number and a simple status check, the citizen cannot tell whether anything happened.

What this means for officers

Raw, unstructured complaints must be read, classified and forwarded by hand before any work starts.

2. Proposed solution

A voice-first AI chatbot **website** that turns a spoken or typed complaint into a routed, trackable ticket:
speak or type → understand → ask only what is missing → confirm → ticket SMD-xxxx at the right office → check status

Need	How Samadhan answers it
No need to know the department	Jev, our decision model, picks the department from the citizen's words; when unsure it asks "is this X or Y?" with questions we wrote
Own language	Hindi and Hinglish text and voice (Sarvam speech-to-text, Groq Whisper fallback) and spoken replies (Sarvam Bulbul)
No need to know the ward	Ward name matched fuzzily to the ward office; GPS is captured for the officer map
Wrong routing	Low confidence or unknown place goes to the district office as <code>needs_review</code> ; officers can reassign, and every correction is logged
Feedback	Complaint number SMD-xxxx; status by typing or saying the number in the chat, or on a status page
Officer workload	Tickets arrive classified, with department, office, a summary and the original audio, plus a review queue and a map

3. Scope of the prototype

- **Departments:** water supply (the original pilot) plus electricity, roads, sanitation and a Human Evaluation desk for unclear cases. **Office data for the last four is DEMO** and labelled so. Ward names are unverified.
- **Coverage:** built for all of Madhya Pradesh. The demo data covers Bhopal (5 demo wards per department plus one district fallback office each). Adding a district means adding data rows, not code.
- **Channels:** website (text, in-browser voice, GPS). WhatsApp, calls and a mobile app are out of scope for now.
- **Information questions** ("how do I get an income certificate?"): a fixed honest reply, a validated `.gov.in` link and a disclaimer. Samadhan does **not** answer government-information questions itself and does **not** read or verify any document.

4. What we deliberately did not do

- No invented answers: the AI cannot create a department, field or office.
- No document handling: documents and government records need the citizen's consent and official integration (for example DigiLocker), which is roadmap work.
- No claim of measured resolution-time improvement. That needs real usage over time.